

Mamdouh Attia

Mobile Software Engineer | Flutter Specialist

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PROFESSIONAL SUMMARY

Flutter Developer with **+3 years** of hands-on experience building cross-platform mobile applications through **Clean Architecture** and **Agile** methodologies. Proficient in **state management** (Bloc/Cubit, Provider), **Firestore**, **REST API integration**, hardware sensors, and **AI features**.

PROFESSIONAL EXPERIENCE

Flutter Developer (Open Source Contributor)

Feb. 2026 – Mar. 2026

Itqan Community (mushaf-imad-flutter)

Remote

- Integrated **Quran.com API** via **OAuth 2.0**, reducing latency by **86%** with custom **in-memory caching**.
- Developed a reactive sync system with **Stream Transformers** for **real-time** verse metadata injection.
- Engineered a **Hybrid Audio Engine** supporting dynamic fallback between local and cloud sources.
- Authored **80+** tests with **Mocktail** for asynchronous repository logic.

Flutter Developer

Sep. 2023 – Present

Freelance

Remote

- Developed a **stress monitoring app** connecting to **ESP8266** microcontroller via **Wi-Fi** with **real-time** sensor reading.
- Enhanced existing client app by adding **Firestore authentication**, **push notifications** & **QR code scanning**.
- Integrated **Blynk API** to enable **real-time** connection with **hardware sensors** in recycle bins.
- Utilized **MVC Architecture** with **Provider** state management.

Flutter Developer

Sep. 2023 – Jun. 2024

PrimaLab

New Damietta, Egypt | Hybrid

- Developed **2 client Flutter applications** (VolunTeen and Check-It) handling all aspects of the **SDLC**.
- Built volunteering app targeting teenagers, featuring **responsive 30–40 screens** with **AI chatbot integration**.
- Developed mobile solution to verify product prices with **barcode scanning** and **real-time price comparison**.
- Integrated **Firestore services**, **GPT-3 API**, **in-app video player**, and third-party services.
- Delivered **50+** hours of hands-on **Flutter training sessions** covering **Dart fundamentals**, **OOP**, **UI development**, layout, widgets, and navigation with practical exercises and project-based learning.

Flutter Developer

Jun. 2023 – Sep. 2023

Orange Digital Center Software House

Cairo- Egypt | Hybrid

- Worked with **agile teams** on **2 internal apps** & **1 hackathon E-wallet app**, meeting tight **sprint deadlines**.
- Translated **Figma designs** into **Flutter UI** and used **MVVM architecture** and **Bloc Cubit**.

EDUCATION

Cairo University - Faculty of Engineering

Cairo, Egypt

Bachelor of Engineering - Computer Engineering Department

2019 – 2024

- Grade: Very Good with Honors
- Graduation Project Grade: Excellent

PROJECTS

Mushaf-IMAD Package | Dart, REST API, OAuth2, Streams, Mocktail | [PR #27](#) | [Demo](#) Feb. 2026 – Mar. 2026

- Integrated global cloud audio streaming with **Single-Trip** optimizations, reducing latency by **86%**.
- Designed a **hybrid timing hierarchy** (RAM → Local → API) ensuring **zero-latency** playback sync.
- Automated testing for complex async data models with **Mocktail**, achieving **80+** tests successful runs.
- Published **architectural guides** and walkthroughs to ensure **long-term maintainability**.

- EcoSort** | *Flutter, Freelance, Firebase, Blync API, QR, Push Notifications* Apr. 2024 – May 2024
- Enhanced an existing client app by adding **Firestore authentication** and **push notifications**.
 - Integrated **Blync API** to enable **real-time** connection with **3 hardware sensors** in recycle bins.
 - Implemented **QR code scanning** and a **point reward system** for recycling activities.
- Check-It** | *Flutter, PrimaLab, Barcode, Provider, Forms* | [Portfolio](#) Feb. 2024 – Jun. 2024
- Developed a mobile solution to verify **product prices** and combat **price manipulation** in retail.
 - Implemented **barcode scanning, real-time price comparison, and e-commerce style product database**.
 - Enabled users to submit complaints and track their status.
- VolunTeen** | *Flutter, PrimaLab, Provider, Firebase, GPT-3* | [Portfolio](#) | [Demo Link](#) Sep. 2023 – Jan. 2024
- Solo client project: built a **volunteering app** targeting teenagers, featuring **responsive 30–40 screens**.
 - Integrated **AI chatbot (GPT-3), in-app video player, and third-party services**.
 - Implemented **MVC architecture, Firebase (storage), shared preferences, and WhatsApp/URL launcher**.
- HyperMate** | *Flutter, Freelance, Wi-Fi service, ESP8266, GPT-3* | [Portfolio](#) | [Demo Link](#) Nov. 2023 – Dec. 2023
- Solo client project: developed a **stress monitoring app** connecting to **ESP8266** microcontroller via **Wi-Fi**.
 - Displayed **real-time GSR sensor readings** and provided personalized exercise and nutrition suggestions.
 - Integrated a smart **chatbot** for user support and recommendations, and handled the **Arduino firmware** code.
- TabCash** | *Flutter, ODC's Hackathon, Agile, MVC* | [GitHub link](#) | [Presentation link](#) Apr. 2023 – May 2023
- Developed a **fintech app prototype** in a **2-week hackathon**, showcasing secure and easy transactions.
 - Built UI for **home, wallet, statistics, and profile screens**, and presented to Orange Digital Centre's board.
- Reddit Clone** | *Flutter, MVC-S, Agile, Provider, REST APIs* | [GitHub Link](#) | [Demo Link](#) Aug. 2022 – Dec. 2022
- Developed a **Reddit mobile app clone** as part of a **6-member team** in a software house environment.
 - Implemented **drawers, settings, and community creation screens**.
 - Contributed to **unit testing, mock services, documentation, code reviews and backend integration**.
 - Followed **Scrum Agile** methodology with dedicated roles for frontend, backend, DevOps, and QA.

GRADUATION PROJECT

- Predicto** | *A+ | Python, OpenCV, Pytorch, Deep Learning, Desktop, Research* | [Demo video](#) Aug. 2023 – July 2024
- Developed an open-source Python library named '**vPredicto**', specialized in **Video Frames Prediction**.
 - Researched and implemented from scratch the latest **SOTA DL models** in this field (**SimVP - PredNet**).
 - Responsible for **GUI desktop app** development using **PyQt**, integrating the '**vPredicto**' library.

TECHNICAL SKILLS

Programming:	Dart C C++ Python Java
Mobile Development:	Flutter Provider Bloc/Cubit MVC MVVM Clean Architecture Streams Shared Preferences Flutter Web Responsive \$ Adaptive Design Animations
Backend & Services:	Firebase Supabase REST APIs API Integration
Hardware Support:	ESP8266 Wi-Fi Service Hardware Sensors Arduino Firmware
Features:	Push Notifications QR Code & Barcode Scanning AI Integration (GPT-3) In-App Video Player Searching & Filtering
Testing & Quality:	TDD (Unit & Integration Testing) Mocking Documentation DevTools Debugging
Software Engineering:	SDLC SOLID OOP Data Structures Algorithms Design Patterns Clean Code Dependency Injection Asynchronous Programming
Tools & DevOps:	Git/GitHub GitHub Actions GitHub Projects Linux Agile/SCRUM Figma AI-Assisted Development(GitHub Copilot - Stitch for UI design - Antigravity)

CERTIFICATES

Career Essentials in GitHub Professional Certificate GitHub Certificate Link	2026
Machine Learning Specialization Coursera Certificate Link	2024
Flutter Internship Certificate ODC Certificate Link	2023
HCIA-AI Huawei Certificate	2021